الجامعة الدولية للرباط
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Internet of Things

Chapter 1: Introduction

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Introduction

What is the IoT?

The term 'Internet of Things' was coined by Kevin Ashton in 1999, to refer to **connecting the Internet to the physical world** via sensors.

Since then, the term has been used to describe any **device** that **interacts with the physical world** around it, either by gathering data from **sensors**, or providing real-world interactions via **actuators** (devices that do something like turn on a switch or light an LED), generally **connected** to other devices or the Internet.

The IoT is thus an ecosystem of devices that:

1. **Gather** data using sensors
2. **Interact** using actuators
3. **Connect** with other devices and the Internet

Introduction

What are sensors?

Sensors are devices that measure a **physical process** and convert it to an electrical signal.

Example of sensors:

- Temperature
- Humidity
- Light
- Acceleration
- Speed
- Proximity
- …

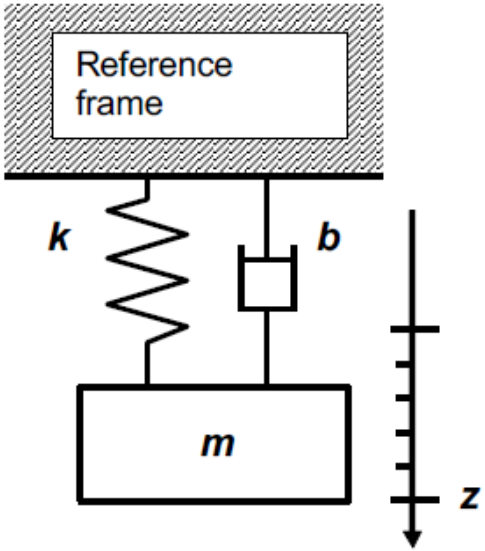
But they can also measure complex signals like:

- Video camera
- Audio signal
- GPS position

Introduction

Example of an accelerometer:

- Many sensors can be modeled as simple mechanical harmonic oscillators
- The figure shows an accelerometer
  - The proof mass (m) is accelerated and displaced
  - The displacement (Δz) of m is measured
  - The acceleration is calculated as $a = k \cdot \Delta z / m$ (The spring constant, k, is known)
  - The highest frequency that this accelerometer can measure is $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

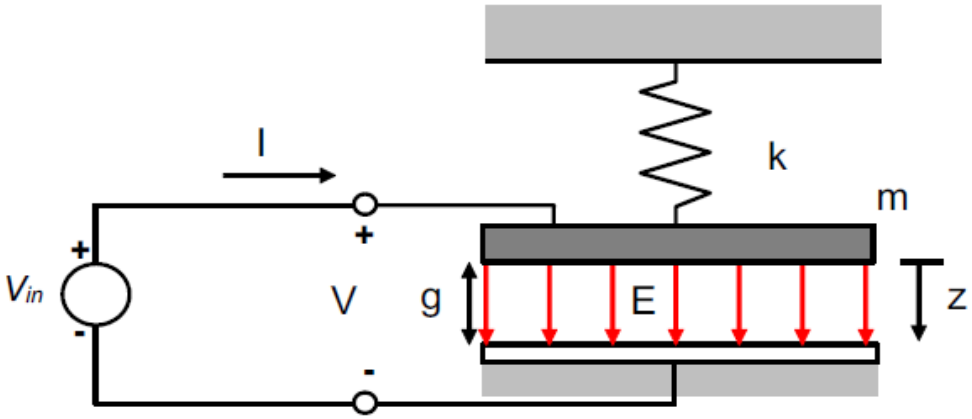


Accelerometer modeled as mechanical harmonic oscillator

Introduction

Example of an accelerometer:

- Most IoT sensors measure changes in electrical properties:
  - Resistance, capacitance, inductance
- Our accelerometer is completed by capacitive measurement of displacement
  - Capacitors are great at measuring short distance with very good absolute accuracy
- Measurements of large displacements are better done by inductors (magnetic fields)



Introduction

What are actuators?

Actuators are devices that convert an electrical signal into a real-world interaction.

Example of actuators:

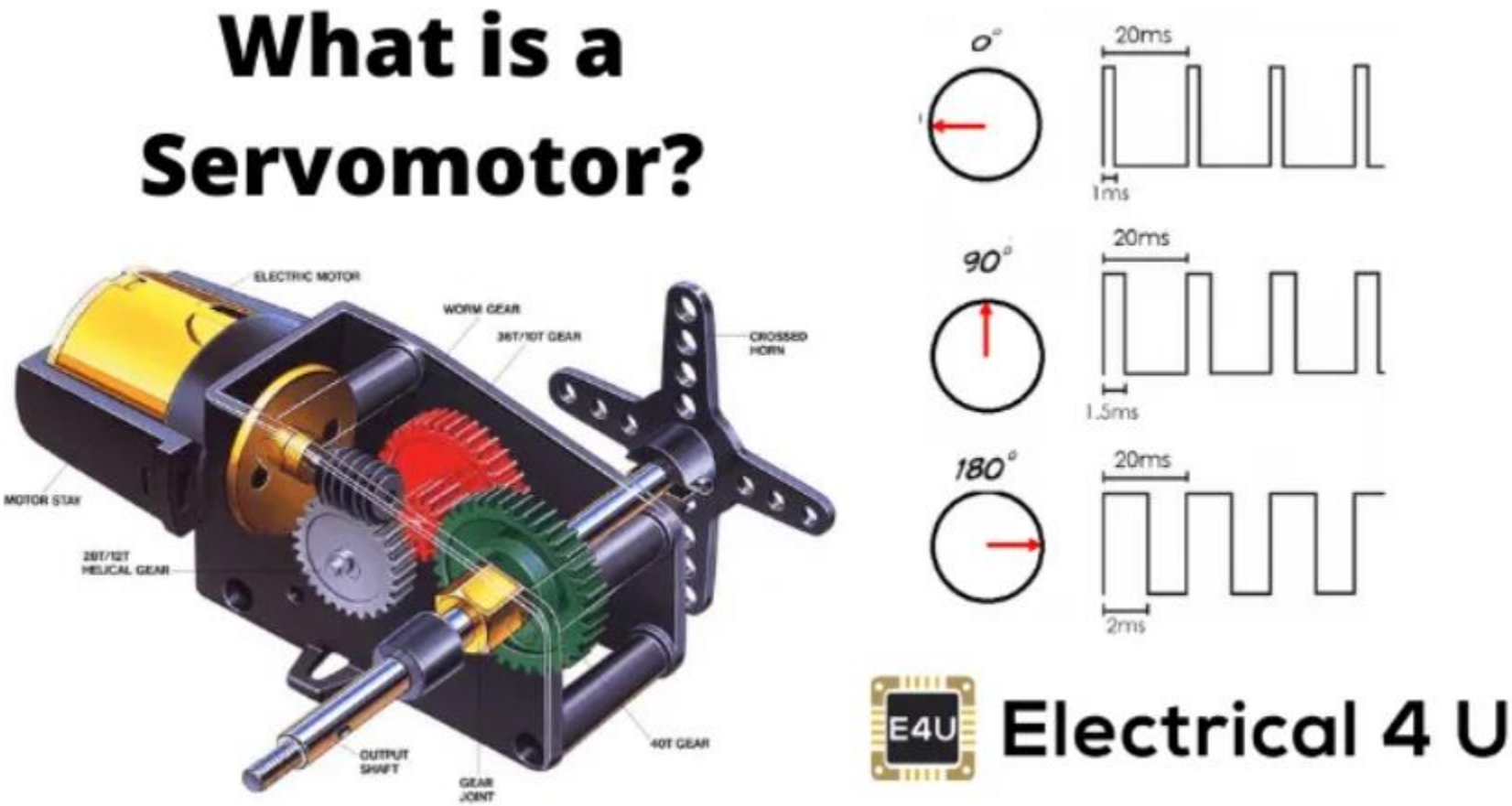
- Switches
- Motors
- LEDs
- Relays
- Buzzers
- …

But there are also actuators that can receive complex signals:

- Displays
- Speakers
- Antennas
- …

Introduction

Example of a servomotor:



Introduction

IoT is more than just the devices

It also includes:

Cloud services:

1. To process the data coming from the IoT devices

2. Make decisions that are dispatched to the devices to command actuators

⇒ Need Internet connectivity

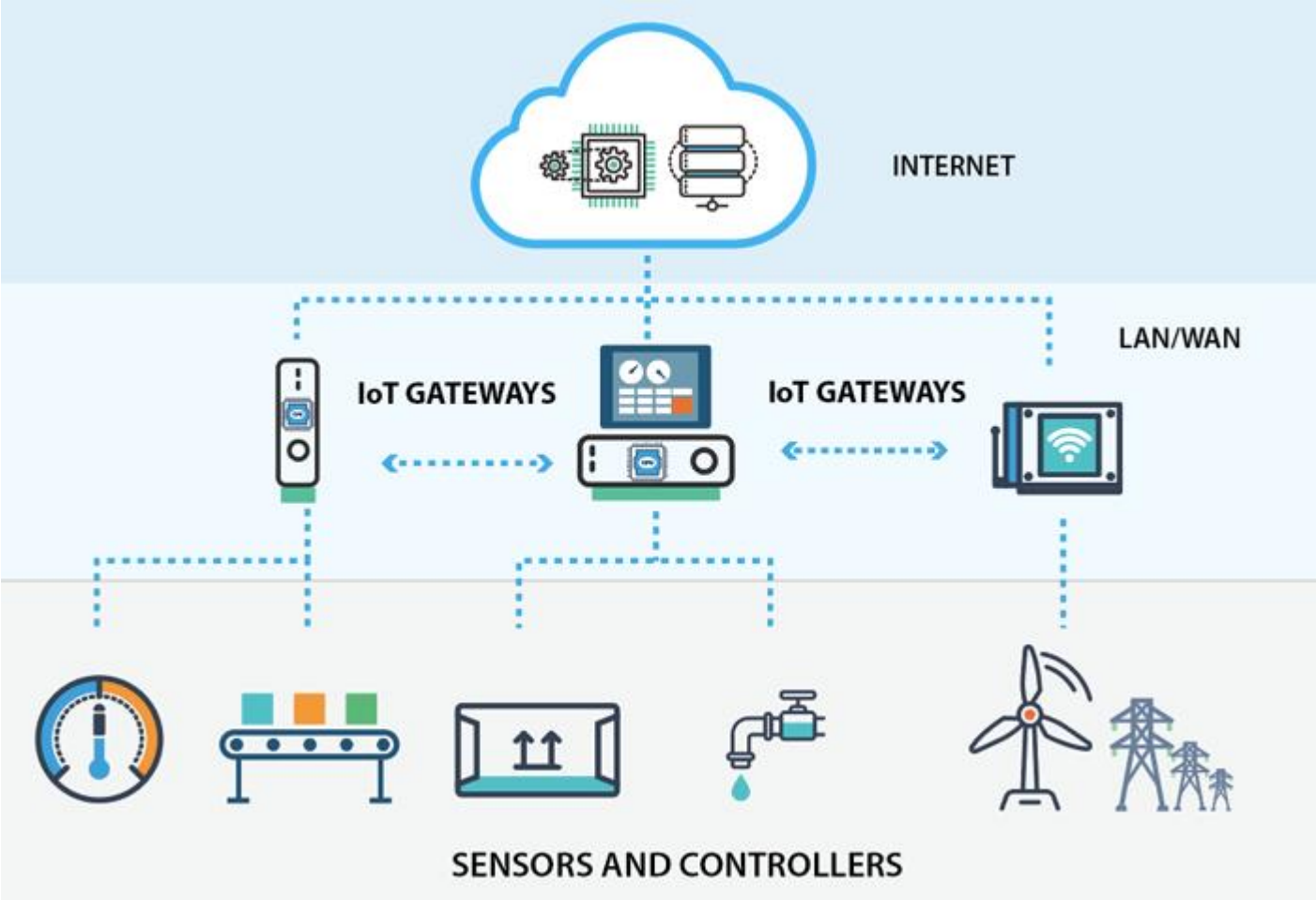
Edge devices:

Can respond to the IoT data and process it **without needing and Internet connection.**

⇒ Example: run AI models that process the IoT device data. The AI model can be trained on the cloud on historical data, and run on the edge to speed up the processing.

Introduction

The IoT gateway



Introduction

What is an IoT gateway?

As the Internet of Things (IoT) expands, businesses find themselves faced with the **challenge of integrating a multitude of devices and protocols**, many with varying power and connectivity requirements. This mix may also include legacy technologies.

IoT Gateways are emerging as an essential component in building a robust IoT and for delivering computational power in edge computing scenarios. Edge computing distributes the load on a system by performing data processing at the data source, or “edge”, rather than relying on a central server for the bulk of the work.

=> IoT gateways are usually the main edge devices in an IoT network

Introduction

What is an IoT gateway?

An IoT Gateway is a solution for **enabling IoT communication**, usually **device - to-device** communications or **device-to-cloud** communications. The gateway is typically a hardware device housing application software that performs essential tasks. At its most basic level, the gateway facilitates the connections between different data sources and destinations.

=> Example: a gateway allows IoT devices using different connectivity technologies (e.g., Bluetooth, Wifi, Zigbee, …) to communicate with each other as well as send data to the cloud.

A simple way to conceive of an IoT Gateway is to compare it to your home or office network router or gateway. Such a gateway facilitates communication between your devices, maintains security and provides an admin interface where you can perform basic functions. An IoT Gateway does this and much more.

Introduction

Example of functions an IoT gateway can perform

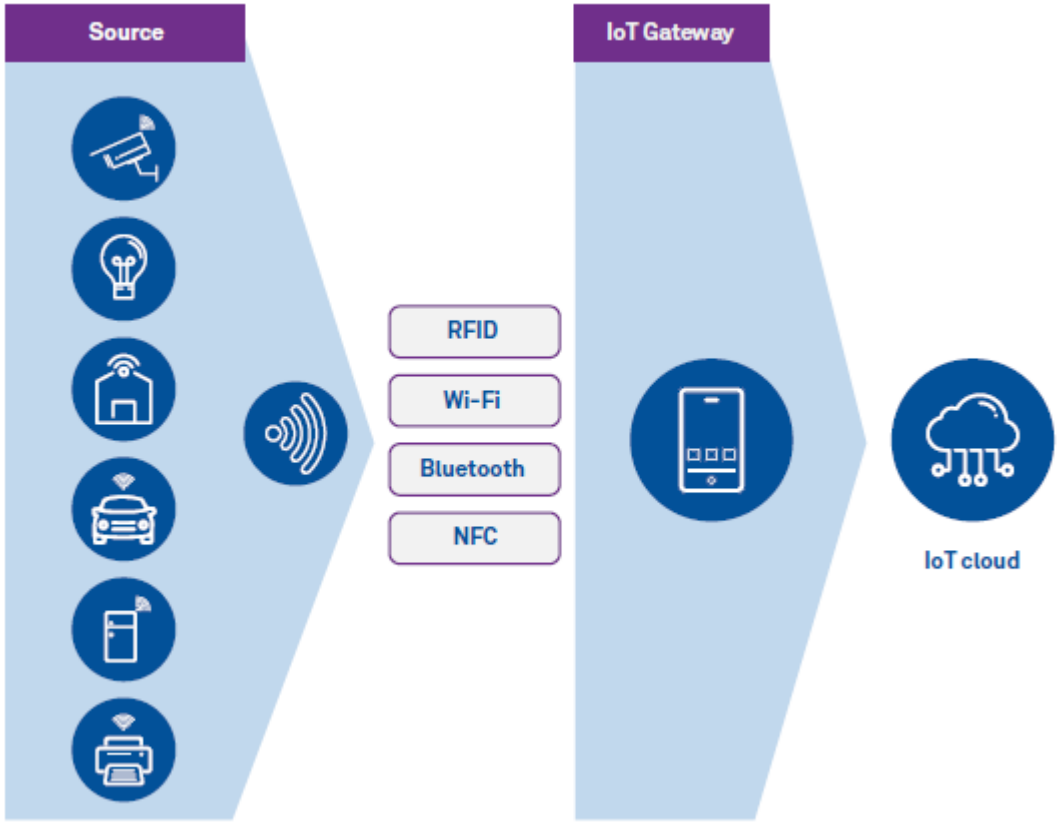
- **Facilitating communication** with legacy or non-internet connected devices
- **Data caching**, buffering and streaming
- **Data pre-processing**, cleansing, filtering and optimization
- Some **data aggregation**
- Device to Device **communications**/M2M
- **Networking** features and hosting live data
- **Data visualization** and basic data analytics via IoT Gateway applications
- Short term **data historian** features
- **Security** – manage user access and network security features
- Device **configuration** management
- System **diagnostics**

Introduction

Example of an IoT gateway: Amazon Alexa

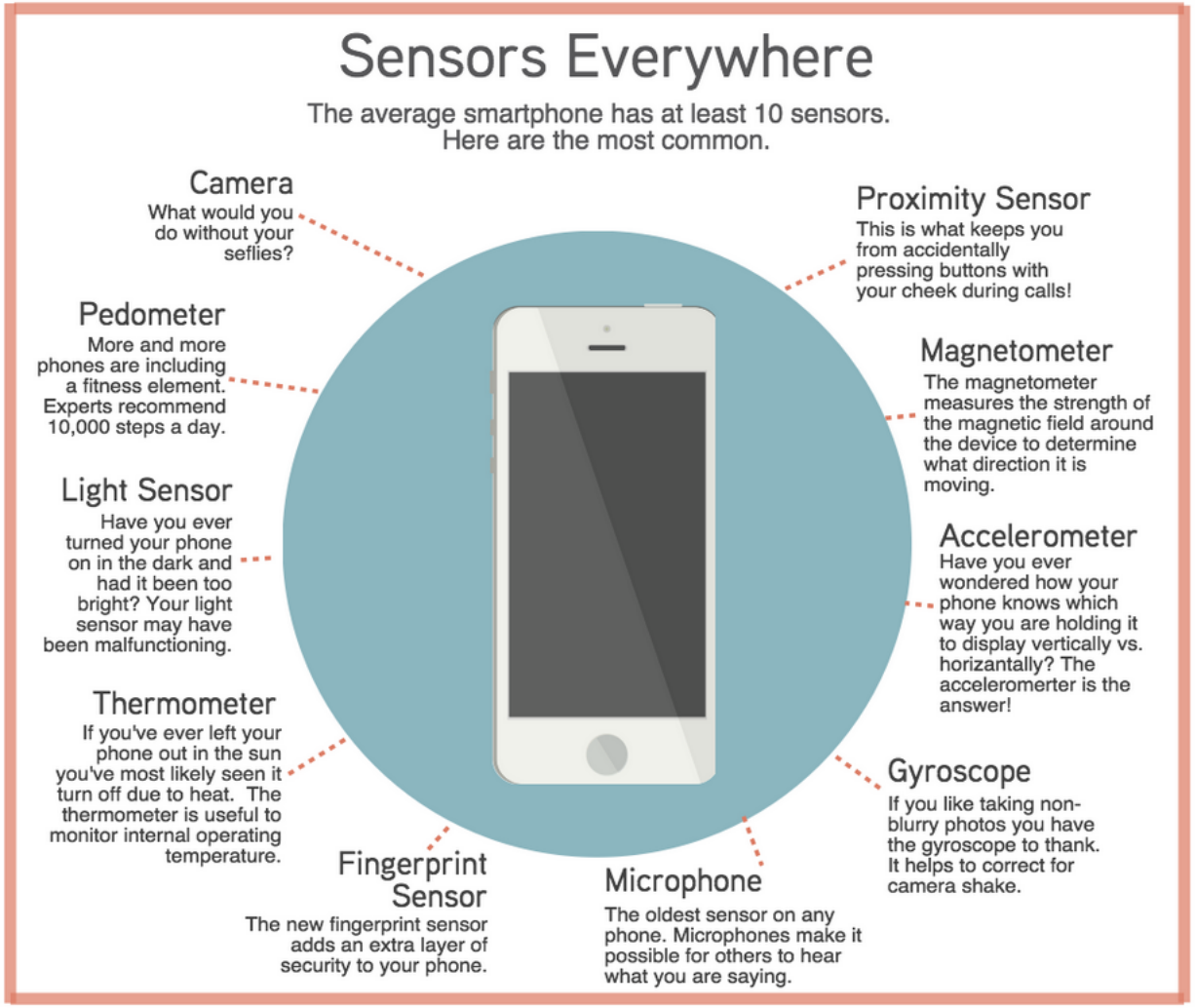
Introduction

Your smartphone can also be an IoT gateway!



Introduction

Your smartphone can also be a normal IoT device!



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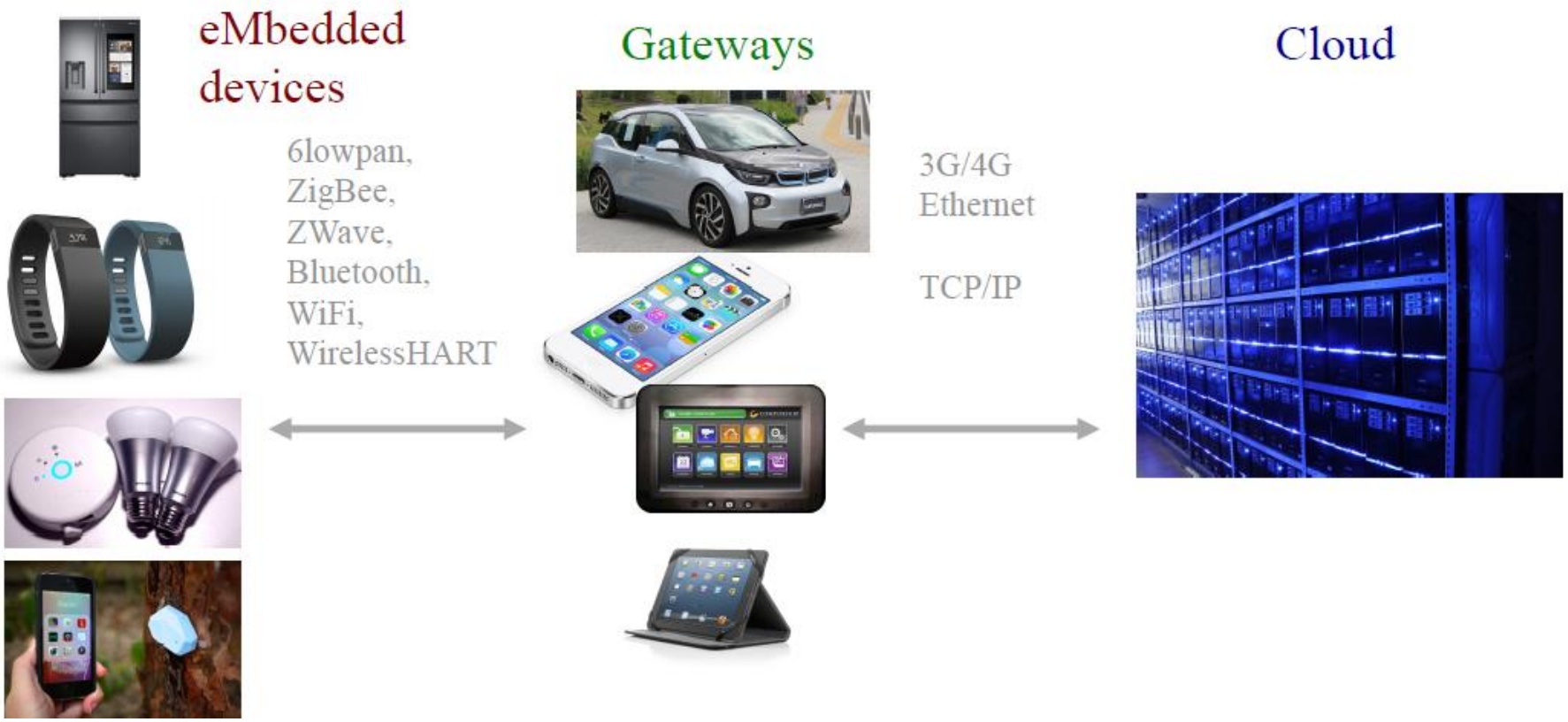
IoT MGC architecture:



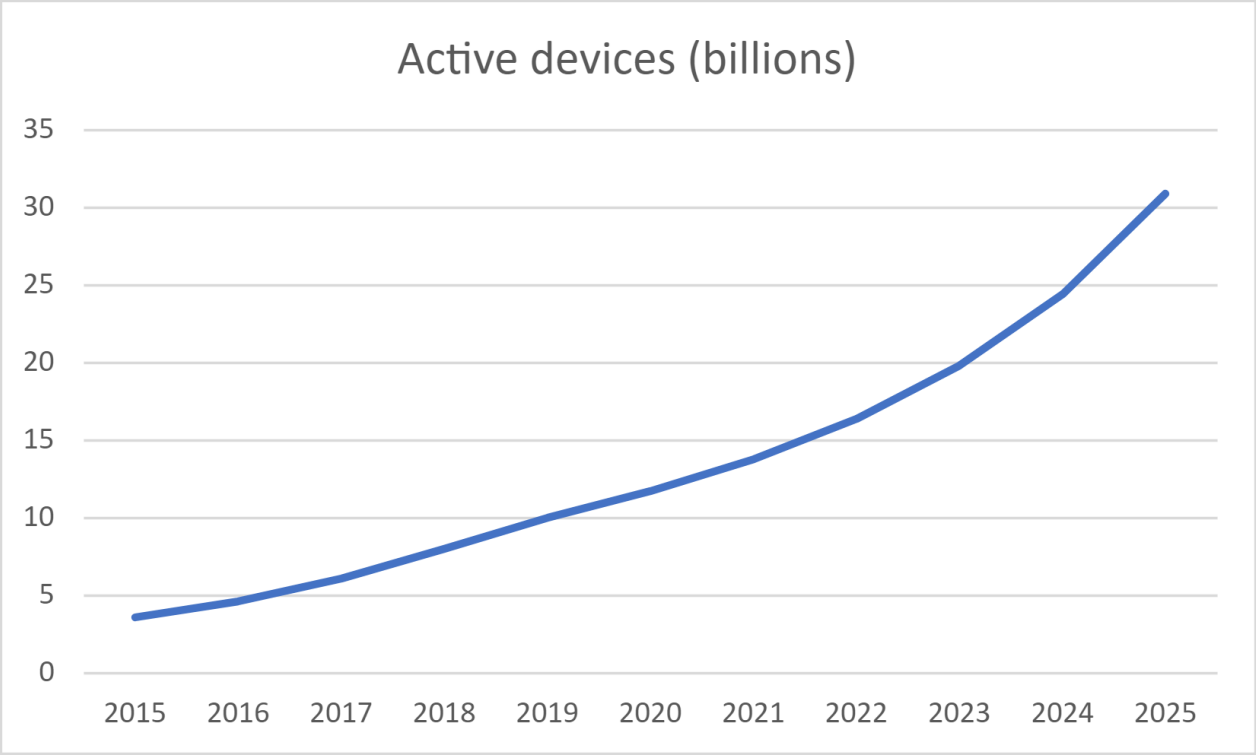
⇒ A Gateway is not necessary but is very useful in many applications

⇒ Some devices can be gateways or normal IoT devices (e.g., smartphone)

Introduction

IoT is a fast growing technology

It is estimated that by the end of 2020, 30 billion IoT devices were deployed and connected to the Internet.

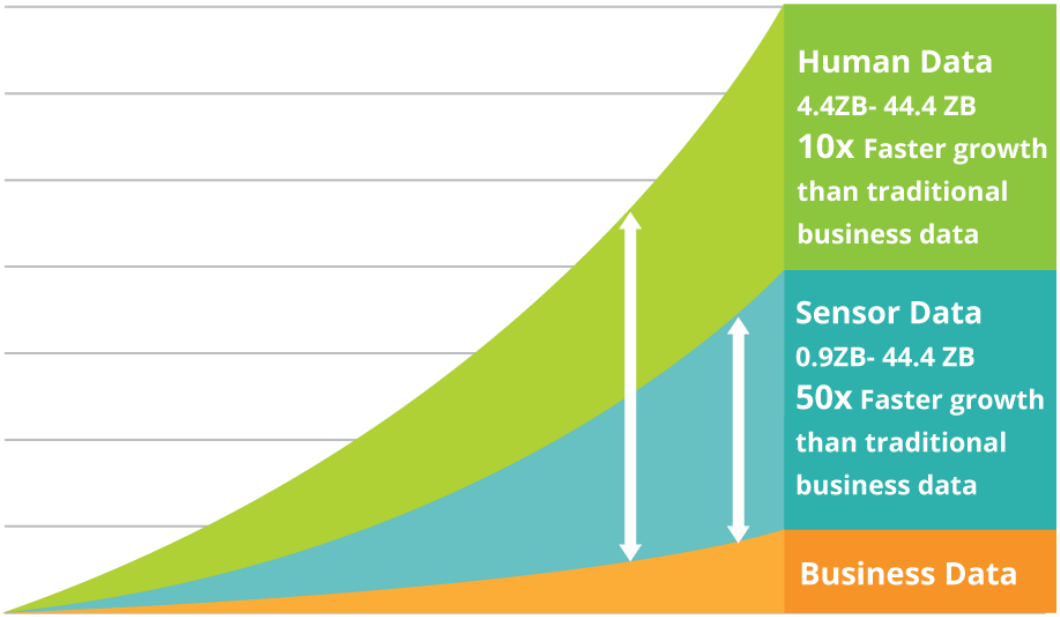


Introduction

The data is the key to IoT's success

Looking to the future, it is estimated that by 2025, IoT devices will be gathering almost 80 zettabytes of data or 80 trillion gigabytes.

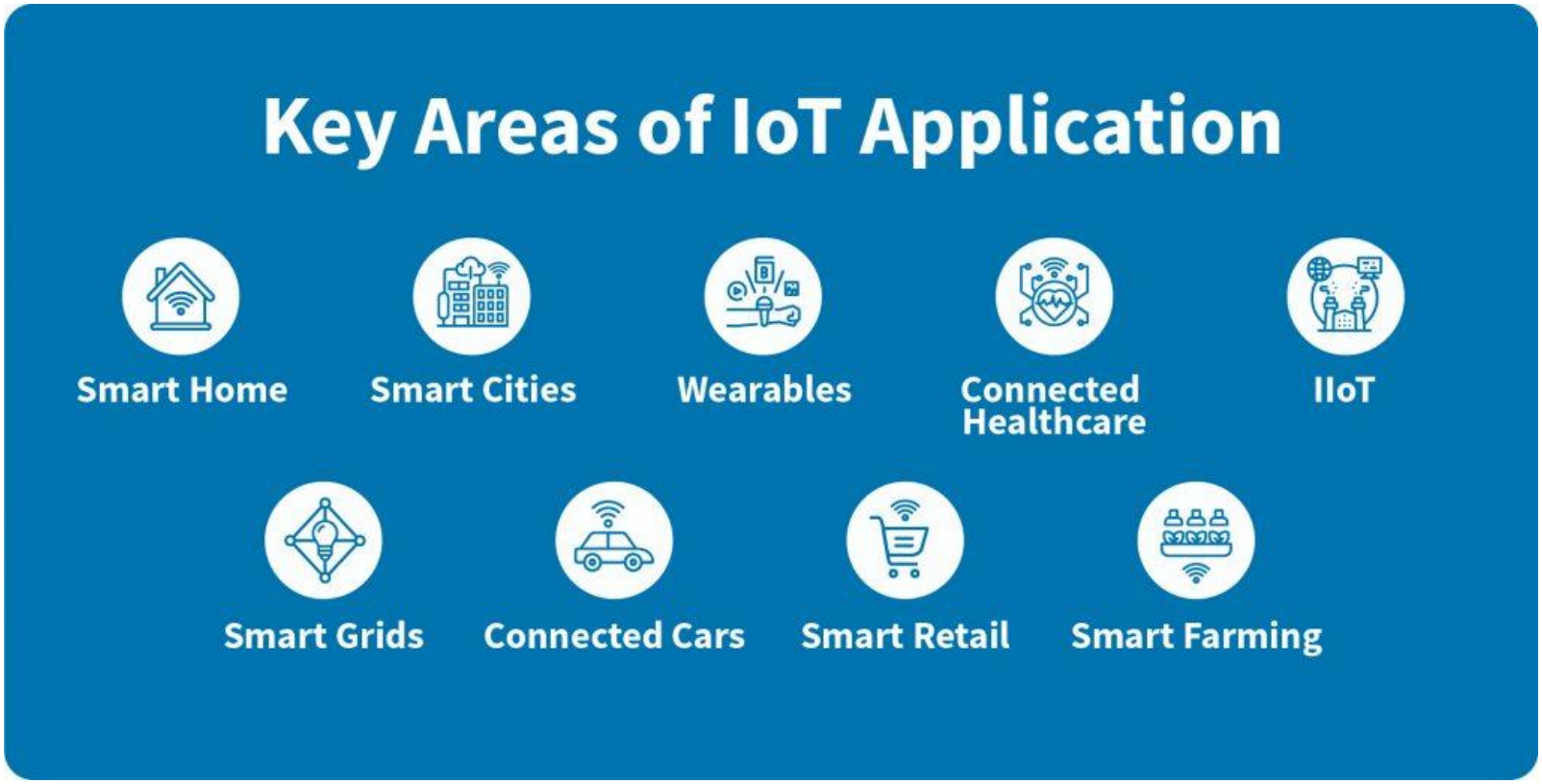
The growth of human and machine-generated data



Source: Inside big data

Introduction

IoT applications



Introduction

IoT applications

Can be classified into 4 main areas:

1. **Consumer IoT**
2. **Commercial IoT**
3. **Industrial IoT**
4. **Infrastructure IoT**

Introduction

Consumer IoT

Consumer IoT refers to IoT devices that **consumers** will **buy and use around the home**. Some of these devices are incredibly useful, such as smart speakers, smart heating systems and robotic vacuum cleaners.

Consumer IoT devices are **empowering people to achieve more in their surroundings**, especially the 1 billion who have a disability. Robotic vacuum cleaners can provide clean floors to people with mobility issues who cannot vacuum themselves, voice-controlled ovens allow people with limited vision or motor control to heat their ovens with only their voice, health monitors can allow patients to monitor chronic conditions themselves with more regular and more detailed updates on their conditions.

These devices are becoming so **ubiquitous** that even young children are using them as part of their daily lives, for example, students doing virtual schooling during the COVID pandemic setting timers on smart home devices to track their schoolwork or alarms to remind them of upcoming class meetings.

=> Includes Smart Home, Wearables, Connected Healthcare

Introduction

Commercial IoT

Commercial IoT covers the use of IoT in the **workplace**.

In an **office** setting, there may be occupancy sensors and motion detectors to manage lighting and heating to only keep the lights and heat off when not needed, reducing cost and carbon emissions. In a factory, IoT devices can monitor for safety hazards such as workers not wearing hard hats or noise that has reached dangerous levels.

In **retail**, IoT devices can measure the temperature of cold storage, alerting the shop owner if a fridge or freezer is outside the required temperature range, or they can monitor items on shelves to direct employees to refill produce that has been sold.

The **transport** industry is relying more and more on IoT to monitor vehicle locations, track on-road mileage for road user charging, track driver hours and break compliance, or notify staff when a vehicle is approaching a depot to prepare for loading or unloading.

=> Includes Smart Retail, Connected Cars

Introduction

Industrial IoT

Industrial IoT, or IIoT, is the use of IoT devices to **control and manage machinery on a large scale**. This covers a wide range of use cases, from factories to digital agriculture.

Factories use IoT devices in many different ways. Machinery can be monitored with multiple sensors to track things like temperature, vibration and rotation speed. This data can then be monitored to allow the machine to be stopped if it goes outside of certain tolerances - it runs too hot and gets shut down for example. This data can also be gathered and analyzed over time to do predictive maintenance, where AI models will look at the data leading up to a failure, and use that to predict other failures before they happen.

Introduction

Industrial IoT

Digital agriculture is important if the planet is to feed the growing population, especially for the 2 billion people in 500 million households that survive on subsistence farming. Digital agriculture can range from a few single digit dollar sensors to massive commercial setups. A farmer can start by monitoring temperatures and using growing degree days to predict when a crop will be ready for harvest. They can connect soil moisture monitoring to automated watering systems to give their plants as much water as is needed, but no more to ensure their crops don't dry out without wasting water. Farmers are even taking it further and using drones, satellite data and AI to monitor crop growth, disease and soil quality over huge areas of farmland.

Introduction

Infrastructure IoT

Infrastructure IoT is monitoring and controlling the local and global infrastructure that people use every day.

Smart Cities are urban areas that use IoT devices to gather data about the city and use that to improve how the city runs. These cities are usually run with collaborations between local governments, academia and local businesses, tracking and managing things varying from transport to parking and pollution. For example, in Copenhagen, Denmark, air pollution is important to the local residents, so it is measured and the data is used to provide information on the cleanest cycling and jogging routes.

Smart power grids allow better analytics of power demand by gathering usage data at the level of individual homes. This data can guide decisions at a country level including where to build new power stations, and at a personal level by giving users insights into how much power they are using, when they are using it, and even suggestions on how to reduce costs, such as charging electric cars at night.

Introduction

Examples of IoT applications

Smart Waste Management

http://ecubelabs.com/case-studies/dublin-airport/

- Go from collecting 840 containers 4 times a day to collecting 80 containers a day.
- Increase waste collection efficiency by 90%.

Introduction

Examples of IoT applications

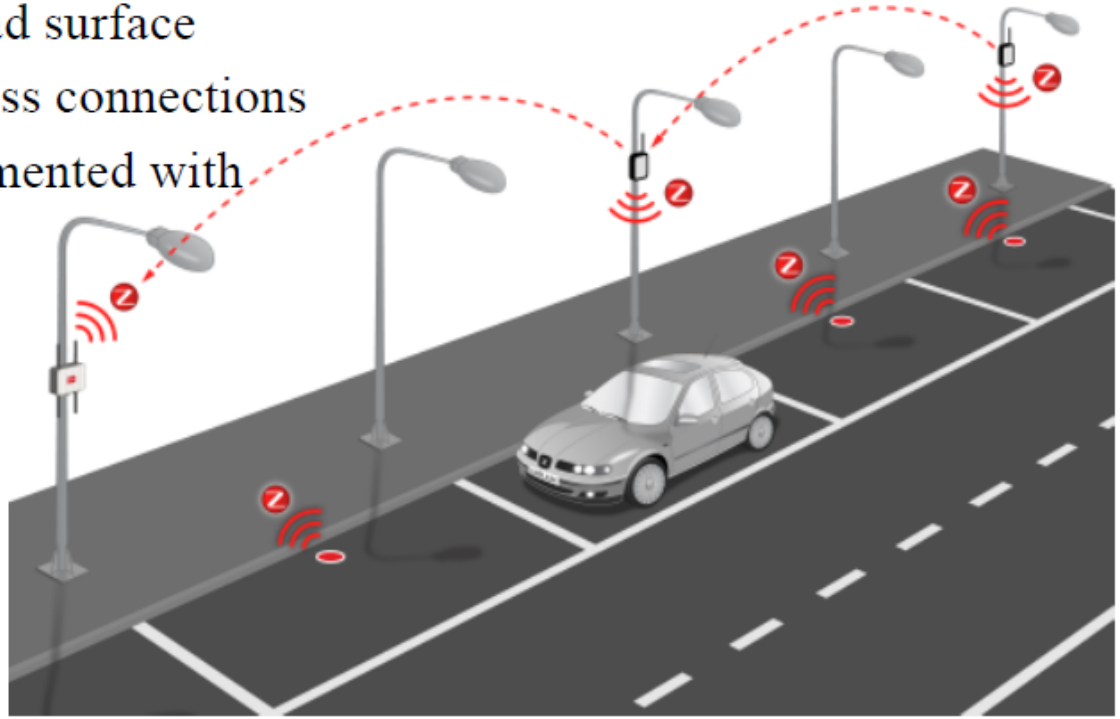
Smart Street Lights

Introduction

Examples of IoT applications

Smart Street Parking

- Infrared- and magnetic-based vehicle detection sensor mounted on the road surface
- Zigbee, LoRaWAN wireless connections
- Mesh networks are implemented with in street lights.
- Apps to direct drivers to empty spaces
- Dynamic parking pricess



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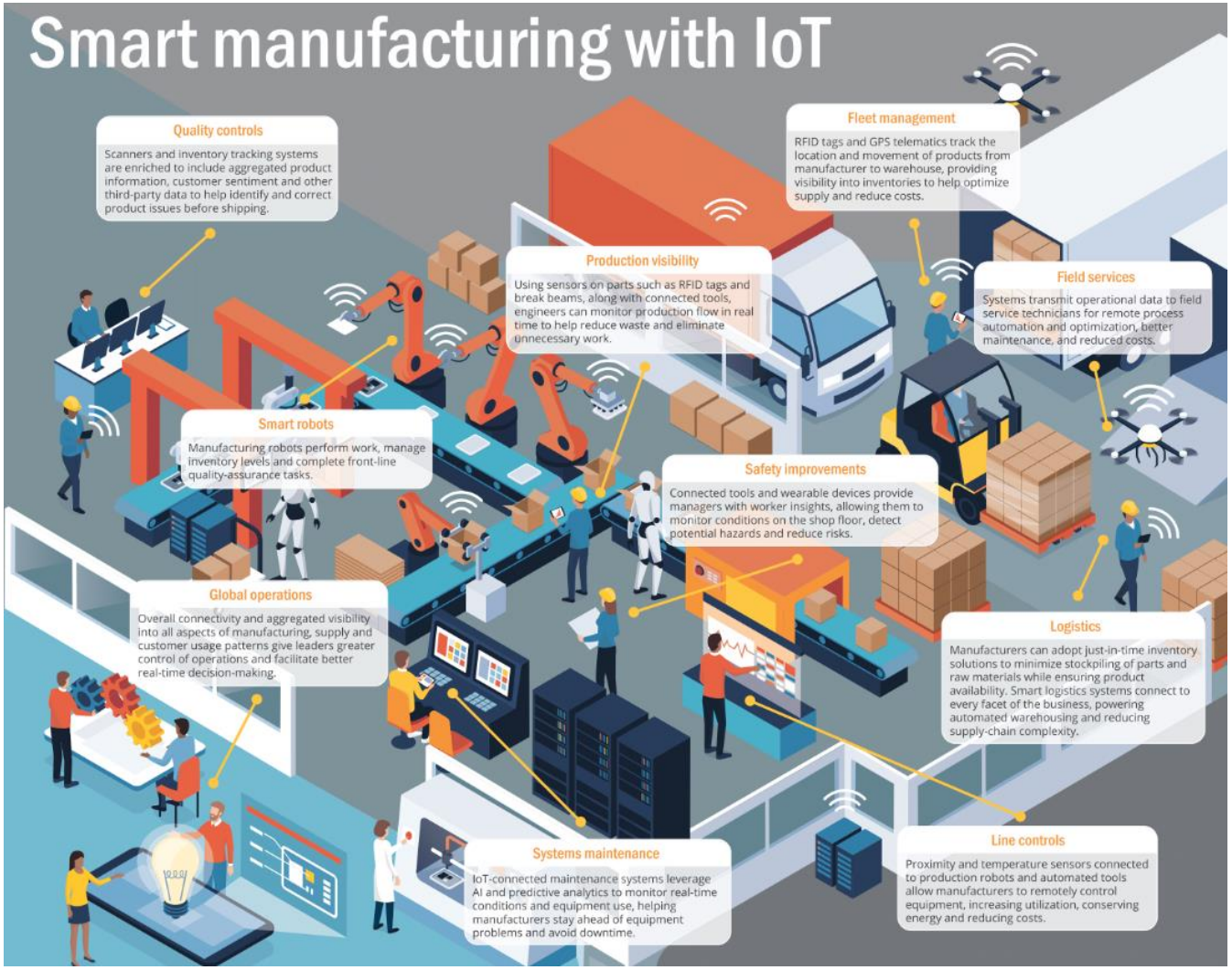
Examples of IoT applications



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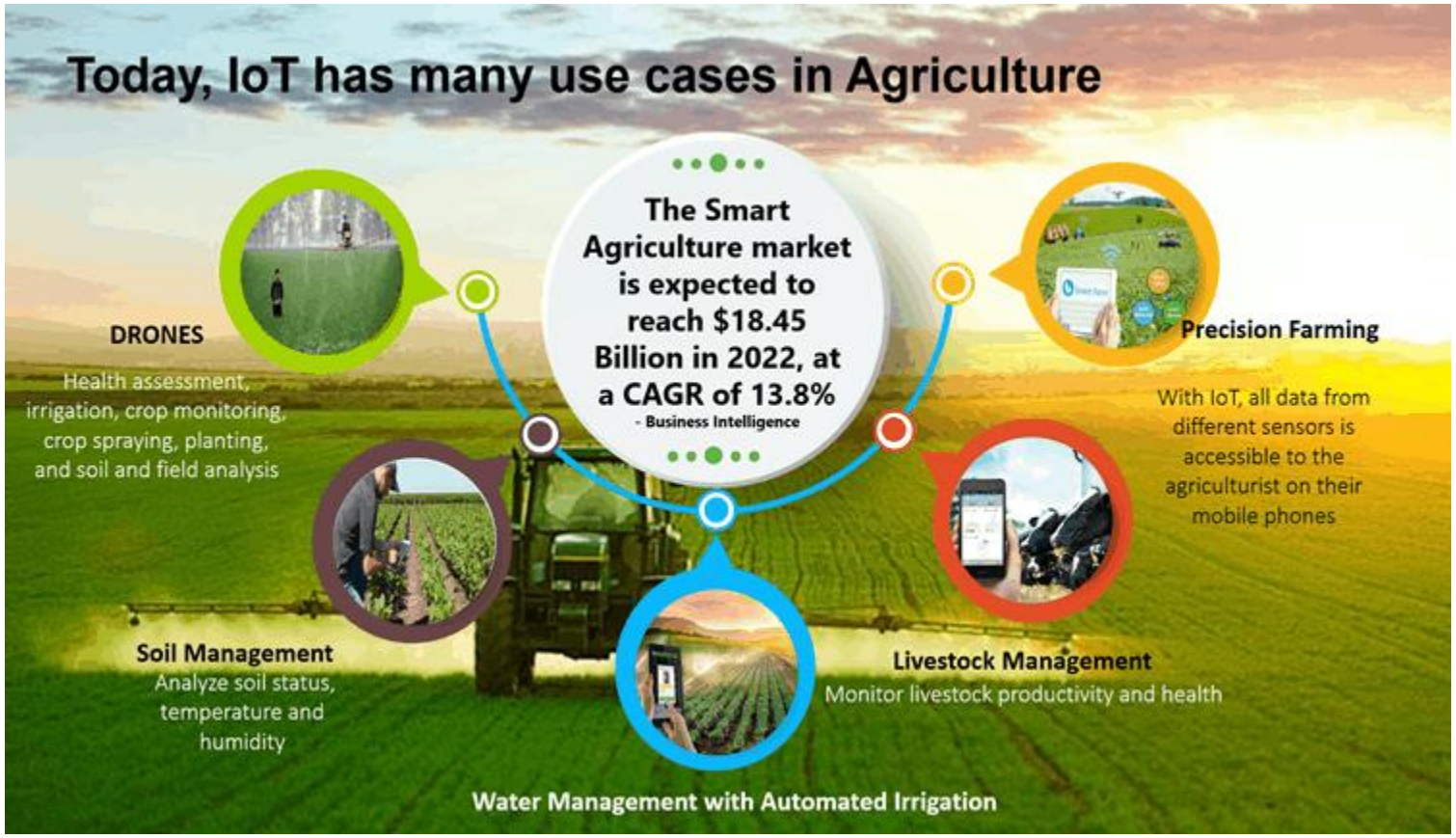
Examples of IoT applications



Introduction

How to create new IoT devices?

The **T** in IoT stands for **Things** - devices that interact with the physical world around them either by **gathering data from sensors** or providing **real-world interactions via actuators**.

Devices for production or commercial use, such as consumer fitness trackers, or industrial machine controllers, are usually **custom-made**. They use **custom circuit boards**, maybe even **custom processors**, designed to meet the **needs of a particular task**, whether that's being small enough to fit on a wrist, or rugged enough to work in a high temperature, high stress or high vibration factory environment.

As a developer either learning about IoT or creating a device prototype, you'll need to start with a **developer kit**. These are **general-purpose IoT devices** designed for developers to use, often with features that you wouldn't have on a production device, such as a **set of external pins** to connect sensors or actuators to, hardware to support debugging, or additional resources that would add unnecessary cost when doing a large manufacturing run.

Introduction

How to create new IoT devices?

The developer kits usually fall into 3 categories:

- **Microcontrollers (MCU)**

- **Single-board computers**

- **Smartphones**: a smartphone can also be considered to be a general-purpose IoT device, with sensors and actuators built-in, with different apps using the sensors and actuators in different ways with different cloud services.

Introduction

Microcontrollers

A microcontroller (also referred to as an MCU, short for microcontroller unit) is a small computer consisting of:

🧠 One or more central processing units (CPUs) - the 'brain' of the microcontroller that runs your program

💾 Memory (RAM and program memory) - where your program, data and variables are stored

🔌 Programmable input/output (I/O) connections - to talk to external peripherals (connected devices) such as sensors and actuators

Microcontrollers are typically low cost computing devices, with average prices for the ones used in custom hardware dropping to around US$0.50, and some devices as cheap as US$0.03. Developer kits can start as low as US$4, with costs rising as you add more features.

Introduction

Microcontrollers

Microcontrollers are designed to be programmed to do a **limited number of very specific tasks**, rather than being general-purpose computers like PCs or Macs. Except for very specific scenarios, you can't connect a monitor, keyboard and mouse and use them for general purpose tasks.

Microcontroller developer kits usually come with **additional sensors and actuators on board.** Most boards will have one or more LEDs you can program, along with other devices such as standard plugs for adding more sensors or actuators using various manufacturers' ecosystems or built-in sensors (usually the most popular ones such as temperature sensors).

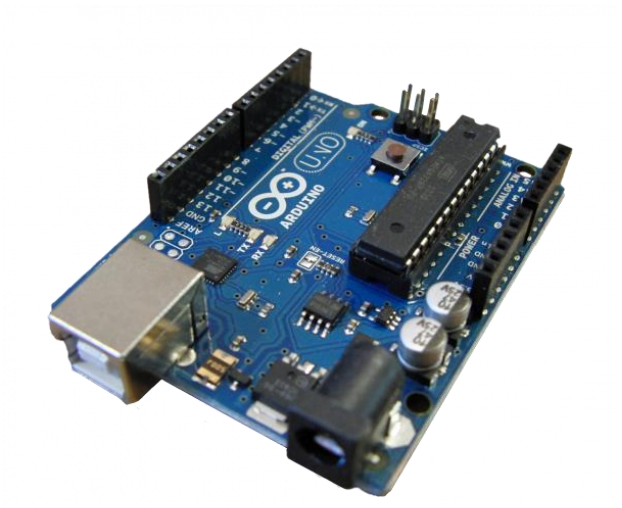
Some microcontrollers have **built-in wireless connectivity** such as Bluetooth or WiFi or have additional microcontrollers on the board to add this connectivity.

Microcontrollers are usually programmed in C/C++.

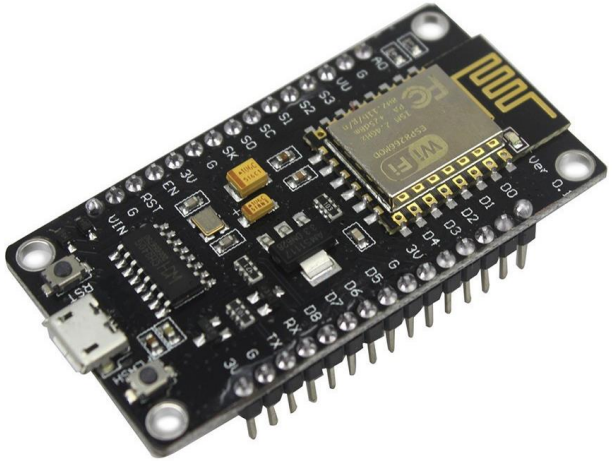
Introduction

Example of Microcontroller dev kits:

- The Arduino UNO:



- The NodeMCU ESP8266: has a WiFi module!



Introduction

Single-Board Computers

A single-board computer is a small computing device that **has all the elements of a complete computer** contained on a single small board. These are devices that have specifications close to a desktop or laptop PC or Mac, run a full operating system, but are small, use less power, and are substantially cheaper.

Like a microcontroller, single-board computers have a **CPU, memory and input/output pins**, but they have additional features such as a **graphics chip** to allow you to connect monitors, **audio outputs**, and **USB ports** to connect keyboards mice and other standard USB devices like webcams or external storage. Programs are stored on **SD cards** or hard drives along with an operating system, instead of a memory chip built into the board.

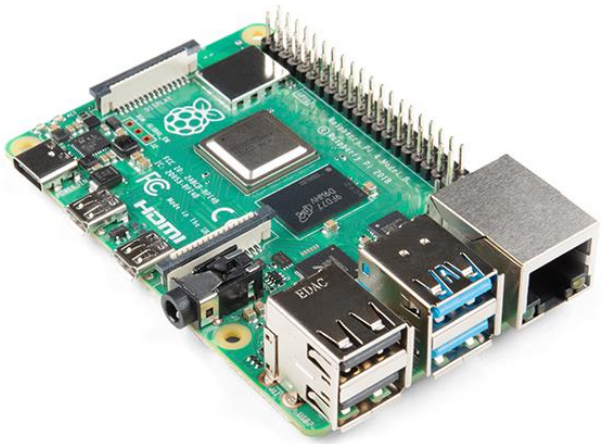
Single-board computers are fully-featured computers, so can be programmed in any language. IoT devices are typically programmed in **Python**.

=> Can be used as a gateway!

Introduction

Example of Single-Board Computers :

- The Raspberry Pi:

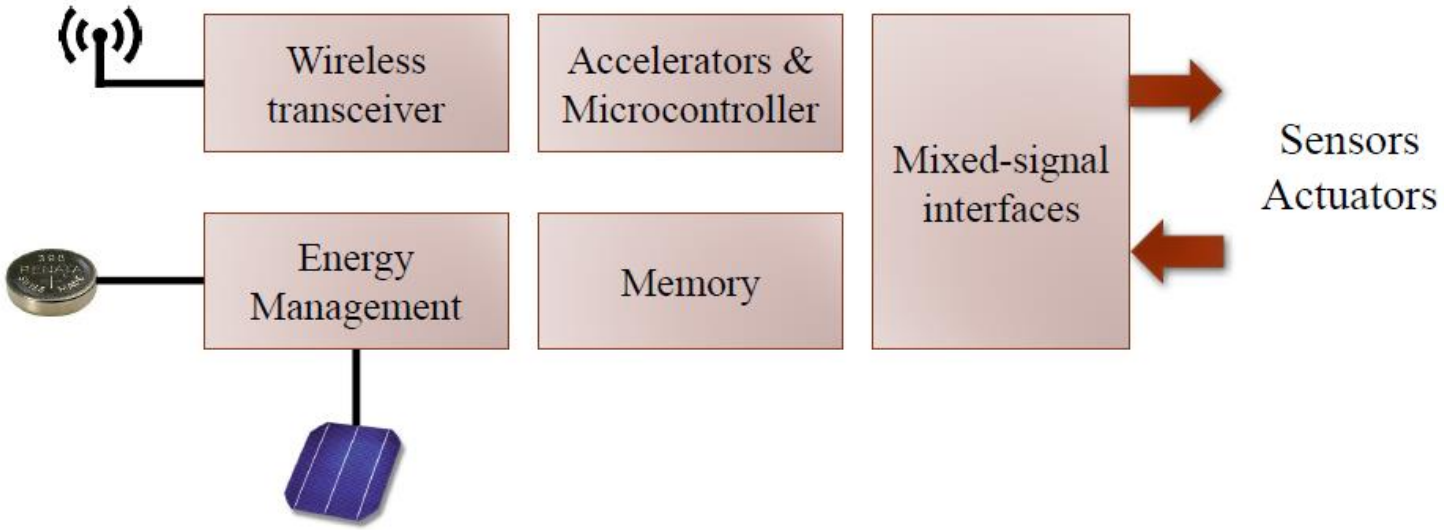


- The Raspberry Pi also has a variant that can be integrated in commercial products: the compute module

Introduction

Production IoT device

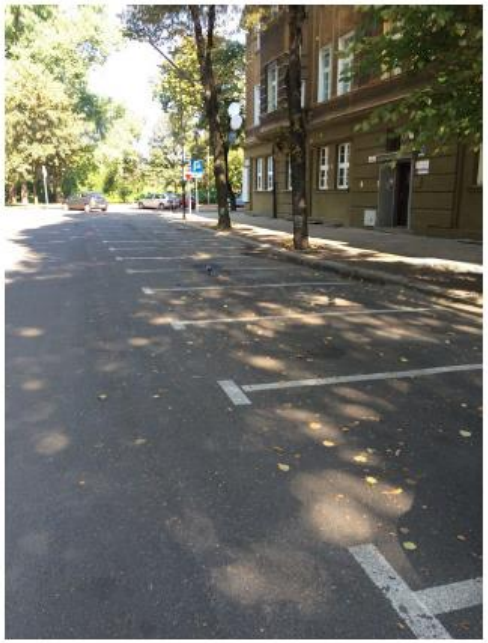
Generic block diagram



Introduction

Key element in production IoT devices: Energy Management

IoT devices may be deployed in hard to reach areas: example, sensors embedded in the street

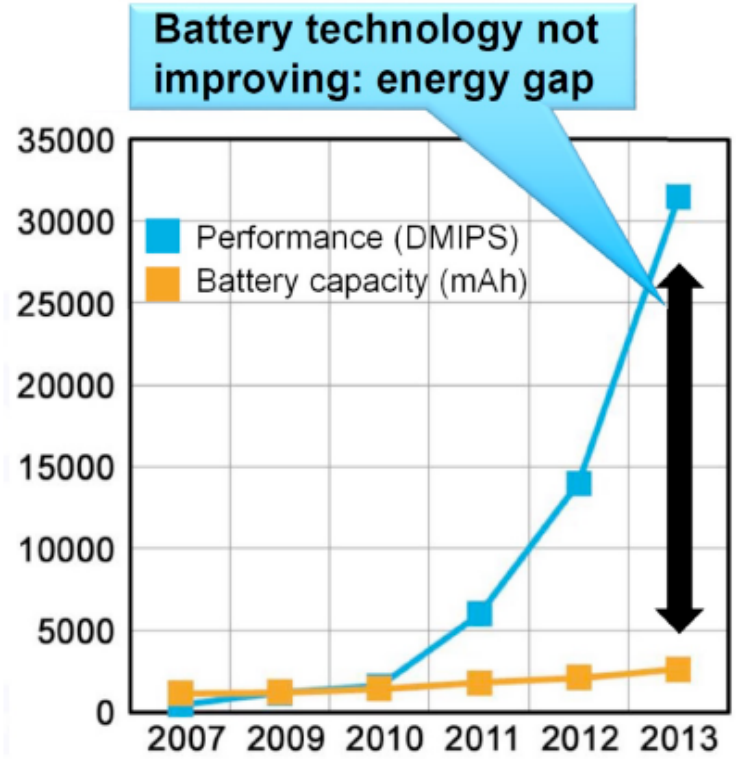


⇒ They cannot be easily charged, or their battery replaced
⇒ They need to be able to use their main energy source for a very long time

Introduction

Key element in production IoT devices: Energy Management

Challenge: battery capacity doesn't evolve very rapidly:

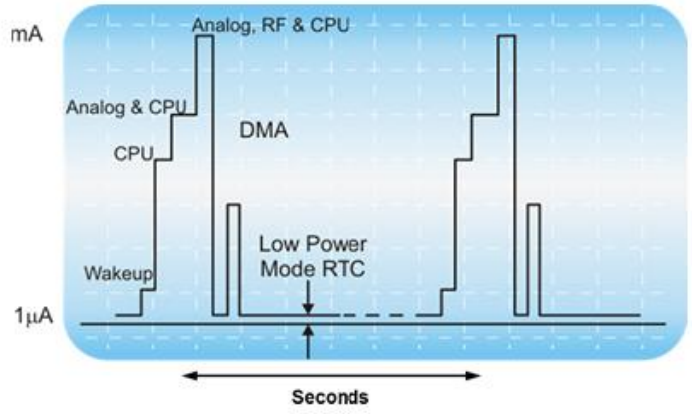


- Semiconductor chip performance tends to improve rapidly
  - See e.g. the blue curve on the left for microprocessors
  - DMIPS = Dhrystone Million Instructions per Second
- Battery capacity improves slowly
  - About 5-8% per year
  - Doubling in ~10 years
- **Key challenge: How to fill this gap via creative circuit & system design?**

Introduction

Energy Management solutions:

Duty cycling:



- IoT transceivers are "mostly" off
- Example: Receive/transmit within 1 ms, then go to sleep for 1 sec

Introduction

Energy Management solutions:

Use lower power wireless technologies:

	Bluetooth LE	Nike+	ZigBee	WIFI
Power (mW)	0.147	0.675	35.7	210
Bits/sec	960	272	192	40M
Energy/bit (nJ)	153	2480	186,000	5.25

- Numbers are a strong function of reach
- Energy per bit tends to be lower for high-data rate links → Invested power amortizes better

Introduction

Energy Management solutions:

Energy harvesting:

Extending battery life (or eliminate batteries!) through harvesting

Source	Limitation	Power Density
Inductive Coupling	Short range (cm), inefficient	$\propto D, Q, 1/d^3$
Far-field RF	Base station (a few m), safety	$\propto 1/d^2$
Solar (Indoor)	Available artificial lighting	10 μW/cm^2
Solar (Outdoor)	Direct sunlight	10,000 μW/cm^2
Vibration	Relatively constant movement	4 μW/cm^2
Thermoelectric	Thermal gradient	25 μW/cm^2

Most popular

Introduction

Summary

- Definition of IoT
- Components of an IoT solution:
  - IoT device
  - Sensors
  - Actuators
  - IoT Gateway
  - IoT Edge Devices
  - Cloud Services
- Examples of IoT applications
- IoT dev kits
- Key challenge for IoT devices: energy management